Analytical Mechanics

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1 Review

Theorem 0.1 (Photoelectric and de Broglie Hypothesis)**.** The energy of a wave is given by

$$E = hf$$

where h is Planck's Constant and f is the frequency. Furthermore, the momentum is given by

$$p = \frac{E}{c} = \frac{h}{\lambda}$$

where $\lambda = \frac{c}{f}$ is the wave length. We may substitute $k = \frac{2\pi}{\lambda}$ and $\omega = 2\pi f$ so where k and ω differ by a factor of c so that

$$E = \hbar\omega, p = \hbar k$$

Furthermore, we may express any wave function as $\psi(x,t) = e^{i(xk-t\omega)}$.

Proof. Take a Galilean transformation with respect to x to get $x = x' - vt$ where ψ has x period λ, so Raise $\psi(x,t) = e^{i(xk-\omega t)}$ to get an invariant. □

Therefore, the momentum operator is $-i\hbar\frac{\partial}{\partial x}$, and the Hamiltonian operator for the single wave

$$\hat{H} = -\frac{\hbar^2}{2m}\nabla^2 + V(r,t)$$

where $\nabla = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial}{\partial z^2}$. Furthermore, we may write $\hbar\omega = i\frac{\partial}{\partial t}$.

Theorem 0.2 (Schrodinger Equation)**.** If Ψ is a wave function, then

$$i\hbar\frac{\partial}{\partial t}\Psi(x,t) = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}\Psi(x,t) + V(x,t)\Psi(x,t)$$

where V is the potential energy of the wave.

Proof. Consider the Hamiltonian operator of a particle

Let $\Psi = e^{xk+t\omega}$. Start with the classical

$$\frac{p^2}{2m} + V(x,t) = E$$

$$\frac{p^2}{2m}\Psi(x,t) + V(x,t)\Psi = hf\Psi(x,t)$$

$$-\frac{\hbar^2}{2m}\frac{\partial}{\partial^2}\Psi(x,t) + V(x,t)\Psi(v,t) = i\frac{\partial}{\partial t}\Psi(x,t)$$

□

Definition 0.3 (Continuous Linear Operator) — We denote